



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science Engineering & Technology
Department of Computer Science and Engineering
Bachelor of Science (B.Sc)
PHY 111 : Physics I
Final Examination
Spring 2024

Batch 2024
Date: 25 - September - 2024

1st Semester
Time: 2 Hour

Answer the following questions

Total Marks: 40

1. Define the following questions (Any 6)

CLO1 6 × 1 = 6

- a) Interference
- b) Average and r.m.s velocity of gas
- c) Malus Law
- d) Heat and Temperature
- e) Fermat's principle
- f) Degree of freedom
- g) Brewster's law
- h) Spherical Aberration with sketch
- i) Astigmatism with sketch

2. Apply your knowledge and proof the following questions (Any 3)

CLO2 5 × 3 = 15

a) Proof that $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ for convex mirror with clear sketch. Where symbol means their usual meanings. Define rectilinear propagation.

b) With clear sketch explain working principle of Nicol prism. And briefly explain polarization by double refractor.

c) For diffraction find the distance from prime maxima to dark as well as bright fringe distance. Briefly sketch and explain intensity vs diffraction angle.

d) Show the rms speed of gas molecule $C_{r.m.s} = \sqrt{\frac{3RT}{M}}$ Where symbol means their usual meanings.

e) From Brewster law proof $r + i = 90^\circ$

3. Apply your knowledge and proof the following questions (Any 1)

CLO3 1 × 7 = 7

a) Derive Smith-Helmholtz equation.

b) Derive Maxwell thermodynamic relation.

4. Analyze the following question and find out the answers (Any 4)

CLO4 $4 \times 3 = 12$

- a) A light beam initially traveling through air hits a diamond with an incident angle of 15° . What is the propagation speed of the light in the diamond? What is the refracted angle? $n_d = 2.717$. (n_d refractive index of diamond)
- b) A small object is placed 20 cm in front of a convex lens with a focal length of 10 cm. The refractive index of the lens is 1.5. What is the height of the image formed by the lens? Object height is 40 cm. Assume the incident light angle is 30° .
- c) A 1.7 kg ice possessing -20°C . If we need to vapor it calculate the heat required to do so.
- d) The speed of light in unknown medium is $2.76 \times 10^8 \text{ m/s}$. Calculate the refractive index of the medium.
- e) If the refractive index of a polarizer is 2.9218. What will be the polarization angle and angle of refraction?
- f) What is the root mean square speed of 6 moles of chlorine gas at 27 degrees Celsius?

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Faculty of Science, Engineering and Technology

1st Semester Final Examination 2024

Course Code: MAT115

Course Title: Mathematics-I (Differential and Integral Calculus)

Final Examination

Semester: Spring 2024

Date: 9/30/2024

Total Time: 2 Hours

Total Marks: 40

Answer any four (4) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

CLOs Marks

CLO1 [1]

CLO2 [4]

CLO3 [5]

Q1. (a) What do you mean by indeterminate forms?

(b) State and prove the L-Hospital theorem.

(c) Determine the following:

(i) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x - 4x}{x^5}$

(ii) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2 \cos x}{x \sin x}$

Q2. (a) State Rolle's theorem and mean value theorem.

CLO2 [3]

(b) Verify Rolle's theorem for the function

$f(x) = x^2 - x$ in the interval $[-1, 1]$.

CLO3 [3]

(b) Discuss the applicability of Mean value theorem for the function

$f(x) = x(x-1)(x-3)$ in the interval $[0, 4]$.

CLO3 [4]

Q3. (a) State and prove the Taylor's theorem and hence expand $\cos(x+h)$.

CLO2, [5]

CLO3

CLO3 [5]

(b) Evaluate the following integrals (any two):

(i) $\int \frac{x^2 \sin^{-1} x^3}{\sqrt{1-x^6}} dx$

(ii) $\int 5 \sin 4x \sin 6x dx$

(iii) $\int x^2 \cos x dx$

Q4. (a) State and prove the fundamental theorem of integral calculus.

CLO2 [4]

(b) Determine the following integrals in terms of partial fractions:

CLO3 [6]

(i) $\int \frac{x^2}{(x+1)(x+2)^2} dx$

(ii) $\int \frac{1}{x^4-1} dx$

Q5. (a) State and prove the Walli's theorem and hence deduce

CLO2, [5]

CLO3

$$\int_0^{\pi/2} y^{1/3} dx$$

where $y = \cos x$.

Q6. (a) Obtain the reduction formula for

CLO2, [5]

$I_n = \int y^n dx$, where $y = \tan x$ and hence determine I_7 .

CLO3

(a) Find the area of the ellipse

CLO3 [4]

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ between the major and minor axes.

(b) Obtain the reduction formula for

CLO2, [6]

$I_n = \int y^n dx$, where $y = \sec x$ and hence determine I_8 .

CLO3

$n^2 + 2n$

10^{n-1}

10^{n-1}



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science Engineering & Technology
Department of Computer Science and Engineering
Bachelor of Science (B.Sc)
EEE 117: Electrical Engineering
Final Examination
Spring 2024

Batch 2024

Date: 02-October-2024

1st Semester

Time: 2 Hour

Answer the following questions (Any 4)

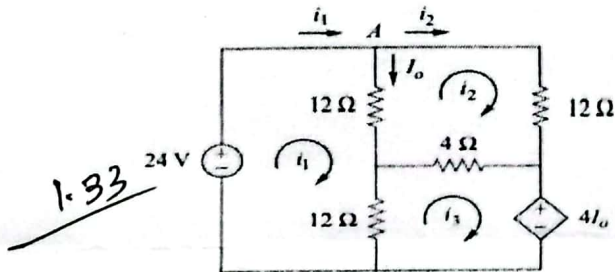
Total Marks: 40

1. (a) Define Mesh analysis
- (b) Show the step response of RC circuit is $v(t) = V_s + (V_0 - V_s)e^{-t/\tau}$
- (c) Use mesh analysis to find the current I_0 in the circuit

CLO1 $1 \times 1 = 1$

CLO2 $4 \times 1 = 4$

CLO3 $5 \times 1 = 5$

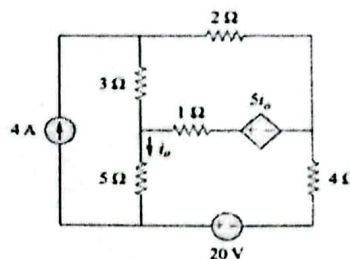


2. (a) Define Thevenin theorem
- (b) Show the step response of RL circuit is $i(t) = i_s + (i_0 - i_s)e^{-t/\tau}$
- (c) Find i_0 the circuit of Figure using superposition

CLO1 $1 \times 1 = 1$

CLO2 $4 \times 1 = 4$

CLO3 $5 \times 1 = 5$



3. (a) Define maximum power transfer condition.
- (b) Derive the condition for maximum power transfer condition in dc network.
- (c) Use Source transformation method to find the current I_0 in the circuit

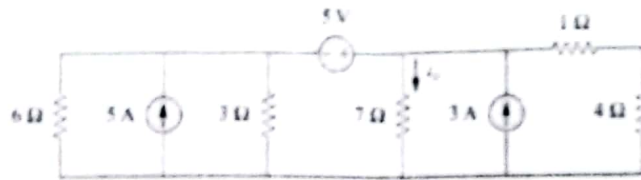
CLO1 $1 \times 1 = 1$

CLO2 $4 \times 1 = 4$

CLO3 $5 \times 1 = 5$

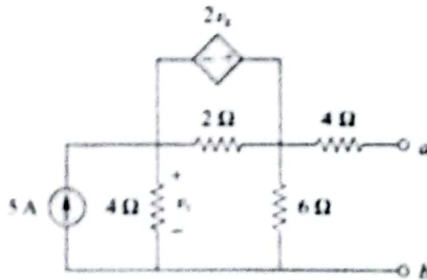
SOURCE TRANSIST

$$2i_1 + 6i_2 = 0$$



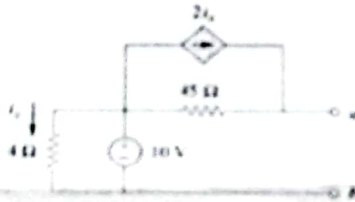
4. (a) Define stored energy formulae of inductor and capacitor
(b) Find the Thevenin equivalent of the circuit using thevenin theorem

CLO1 $2 \times 1 = 2$



CLO3 $4 \times 1 = 4$

- (c) Find the Norton equivalent of the circuit using Norton theorem

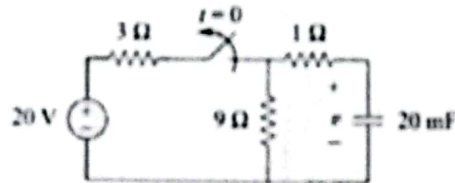


CLO3 $4 \times 1 = 4$

5. (a) Write equivalent formulae of two inductors and two capacitor(separately)
for (i) series (ii) parallel combination .

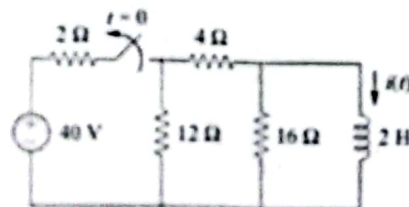
CLO1 $2 \times 1 = 2$

- (b) The switch in the circuit in Fig. has been closed for a long time, and it is opened at $t=0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor



CLO3 $4 \times 1 = 4$

- (c) The switch in the circuit of Fig has been closed for a long time. At the switch is opened. Find $i(t)$ for $t \geq 0$



CLO3 $4 \times 1 = 4$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 117 : Engineering Ethics and Cyber Law

Batch: 26th

Date: 07/10/2024

Final Examination

Semester: Spring 2024

Time: 2 Hours

Total Marks: 40

Answer **five** questions of the followings

5 × 8 = 40

CLOs **Marks**

CLO5 2

CLO5 6

CLO5 3

CLO5 5

CLO5 3

CLO5 5

CLO3 3

CLO5 5

CLO4 2

CLO4 6

CLO3 3

CLO3 2×2.5=5

CLO5 3

CLO4 5

1. (a) What are information warfare and cyber warfare?

(b) Analyze the information war over the internet from current global perspectives.

2. (a) Define a firewall and list the different types of firewalls.

(b) Write the significant uses of firewalls.

3. (a) List the fields targeted by cyber attackers using Denial-of-Service attacks.

(b) Explain the working procedure of Denial-of-Service attacks.

4. (a) Define following terms according to Copyright Act, 2000:

i. Publication

ii. Author

iii. Work

(b) Explain the criteria for determining the first owner of a work according to the Copyright Act, 2000.

5. (a) What are Cyber law and Cyber Crime?

(b) Explain the most common types of cybercrime with appropriate examples.

6. (a) What is patent? Define the types of patent.

(b) Define following terms from the aspect of cyber law

i. Cyber Security

ii. Confidentiality

7. (a) Mention the key objectives of The ICT Act, 2006.

(b) Why Digital Security Act is called as "Draconian" law? Justify this with proper reasons.

ENG 111: Basic English/ English for Technical Communication
Final Examination
Spring 2024

5 October 2024

1st Semester
Time: 2 hours

Total Marks: 40

CL O	Questions	Level	Marks
<u>Part A</u>			
CLO 5	1. Imagine Jamil and you are two students of CSE/EEE/Math discussing the future career. Write a natural and realistic dialogue where Jamil and you will discuss: A. Why did both of you take admission to the CSE/EEE/Math department? B. What career opportunities can you get from your subject? C. What are both of your future career plans? D. How can your future careers contribute to the development of Bangladesh?	Apply	10
CLO 4	2. Suppose you are a fresher of the department of CSE/EEE/Math, seeking for a job in you subject related multinational company. Now write a convincing CVshowcasing your details.	Apply	10
CLO 4	3. Assume you are Mr. Ridwan Rabbi. You have completed you bachelor and master's degree from the department of CSE/EEE/ Math of Hamdard University Bangladesh. Now, write a cover letter to the registrar of Hamdard University Bangladesh for the post of lecturer in the department of CSE/EEE/ Math.	Apply	10
Part B			
CLO 1	4. Rewrite the following sentences using the instructions provided in parentheses: a. Karim has to look after his poor parents. (Negative) b. Kanika is not slicing the potatoes. (Passive) c. Nobody appreciates a dishonest man in this world. (Interrogative) d. You should drink plenty of water. (Imperative) e. Neha wanted to help her father. (Interrogative) f. Rezaul must not be absent in the class. (Imperative) g. He is the only child of his parents. (Negative) h. Were you not good enough for the post? (Assertive/ Affirmative) i. You should drink plenty of water. (Imperative) j. Seema has crafted this showpiece.(Passive)	Apply	0.5X 10=5

CLO 1. In the following sentences, identify and correct any subject-verb agreement errors. If no error found then just write 'correct'.

Apply 0.5X
10=5

- a. Physics is my favorite Subject.
- b. The Poor is suffering from hunger.
- c. People of Venuzuela was not bad for hospitality.
- d. Each of the boys think of missing class.
- e. Rony as well as his father went to the zoo.
- f. Tea and biscuit is my favorite breakfast.
- g. The principle and manager are going home.
- h. One of the most beautiful cities around the world are Kabul.
- i. Elly along with her parents will be going home.
- j. There is five groups of completing the task.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 26th

Semester: Spring 2024

Date: 29/04/2024

Final Examination

Time: 2 hours

Total Marks: 40

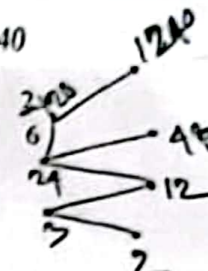
Answer any five questions from the followings.

1. (a) State the process to determine Hasse diagram from Partial Ordering.
(b) Let $A = \{2, 3, 12, 24, 48, 120, 240\}$ be a set. Find out $R = \{(a, b) \mid a \text{ divides } b\}$. Draw the partial ordering diagram for R and finalize the Hasse diagram from the partial ordering diagram, also find out the maximal and minimal.

5x8 = 40

CLO 4

CLO 4



2. (a) How many combinations are there to select 6 players from a 19-member tennis team of CSE Department to make a match with Mathematics Department?
(b) Write the construction procedure of Euler circuit.
(c) Prove that the set of even integer forms a group structure.

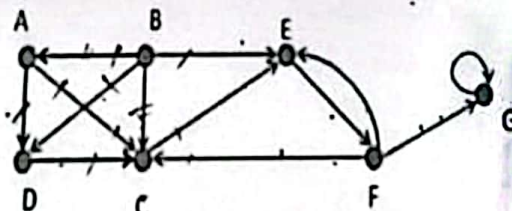
CLO 1 2

CLO 1 3

CLO 1 3

$$H_n = (n-1) \\ = n - n + 1 \\ H_1$$

3. (a) Write the corollaries for Planar graph.
(b) Consider the following graph:



i)	Find the properties with reason, adjacency matrix and adjacency list for the given di-graph:
ii)	Determine degree, in-degree and out-degree of each vertex for the given di-graph.

Handwritten calculations for the graph problem:

$$R = e - v + 2 \\ = 5 - 7 + 2 \\ = 0$$

4. (a) What are the objectives of Pizza-cut problem?
(b) Determine the general equation for the Tower of Hanoi Problem.
(c) What is meant by Recurrence relation? Find the first five terms of the sequence defined by the recurrence relation and initial condition given below:

CLO 4 1

CLO 4 3

CLO 4 4

$$a_n = na_{n-1} + n^2a_{n-2}, a_0 = 1, a_1 = 1.$$

5. (a) State the Dirac's and Ore's theorem for graph.
(b) How many edges are there in a graph with 20 vertices each of degree 5?
(c) Prove that $K_{3,3}$ is not a planar graph.

CLO 4 2

CLO 4 2

CLO 4 4

6. (a) Write the merits and demeris of adjacency matrix and adjacency list of representing a graph.
(b) State the properties of a tree.

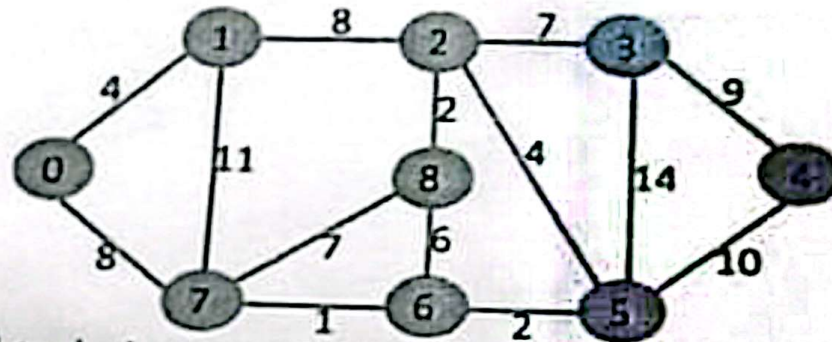
CLO 4 4

CLO 4 4



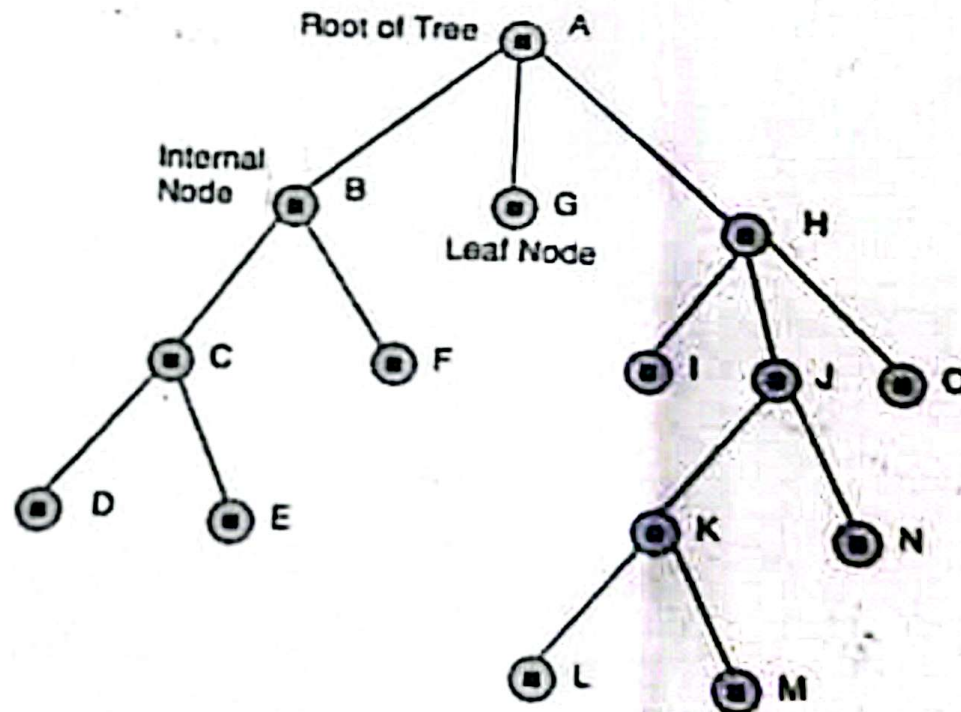
- (a) Find the minimum spanning tree for the following weighted graph by CLO 4 4
using

- (i) Prim's algorithm
(ii) Kruskal's algorithm



- (b) Briefly describe tree traversal techniques and find the order of the elements after traversing the following tree:

CLO 4 4



Faculty of Science, Engineering and Technology

Department of CSE

CSE 111: Basic Computer Skills

Batch: 26th (Section 2)

Semester: Spring 2024

Time: 40 minutes

Date: 17/09/2024

Final Examination

Total Marks: 40

Answer all the questions:	Marks
(a) What does computer mean?	2
(b) Describe the basic organization of a computer system with figure.	5
(c) Write short notes on Primary storage and Secondary storage system.	5
(d) What's the jobs of Central Processing Unit (CPU), Control Unit (CU), Cache memory and Arithmetic Logic Unit (ALU)?	8
(e) Write the differences between RAM and ROM.	5
(f) Convert the following numbers into mentioned forms: (i) $(ABC)_{16} = (?)_{10}$ (ii) $(11011011)_2 = (?)_{10}$	5
(g) Describe how do we keep our computers safe from virus?	5
(h) Write the differences between internet and intranet	5